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ABSTRACT

Handwritten digit recognition is one of the fundamental problems in the fields of machine learning, pattern recognition, and artificial intelligence. It focuses on identifying and classifying numerical digits written by humans in the form of images. This problem has practical importance in many real-world applications such as postal mail sorting, bank check processing, automated form reading, and document digitization. Due to variations in individual handwriting styles, recognizing handwritten digits accurately is a challenging task for computer systems.

In this project, a digit recognition system is developed using an Artificial Neural Network (ANN). Artificial Neural Networks are computational models inspired by the structure and functioning of the human brain. They consist of interconnected processing units called neurons that are capable of learning patterns from data through training. Neural networks are widely used in machine learning tasks such as image classification, speech recognition, and natural language processing.

The model in this project is trained using the MNIST dataset, which is one of the most commonly used benchmark datasets for handwritten digit recognition. The dataset contains 70,000 grayscale images of handwritten digits ranging from 0 to 9. Each image has a resolution of 28 × 28 pixels, making it suitable for training machine learning models. Out of the total dataset, 60,000 images are used for training the neural network while 10,000 images are used for testing the model's performance.

Before training the model, the dataset undergoes preprocessing where pixel values are normalized to improve the learning efficiency of the neural network. The ANN model is constructed using multiple layers including an input layer, hidden layers with activation functions, and an output layer that classifies the digits into ten possible categories. The model is implemented using Python programming language with the help of TensorFlow and Keras libraries in the Google Colab environment.

During the training phase, the neural network learns important features from the handwritten digit images by adjusting its weights and biases through an optimization process. The performance of the model is then evaluated using test data that was not seen during training. The results obtained from the experiment show that the ANN model can successfully classify handwritten digits with high accuracy.

INTRODUCTION

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make decisions without being explicitly programmed. Instead of following fixed instructions, machine learning algorithms improve their performance by learning from examples. These techniques are widely used in modern applications such as recommendation systems, speech recognition, image classification, and autonomous systems.

Artificial Neural Networks (ANN) are one of the most important models used in machine learning. They are inspired by the structure of the human brain and consist of interconnected processing units called neurons. These neurons are organized into different layers, including an input layer, one or more hidden layers, and an output layer. During training, the network learns patterns from data by adjusting internal parameters called weights and biases. Through this learning process, the network becomes capable of making accurate predictions on new and unseen data.

One of the most common applications of Artificial Neural Networks is image recognition. Image recognition involves identifying objects or patterns present in digital images. A widely studied problem in this domain is handwritten digit recognition. This task involves recognizing numerical digits written by different individuals. Since handwriting styles vary from person to person, designing a system that accurately recognizes digits can be challenging.

Handwritten digit recognition has several practical applications in real-world systems. For example, banks use automated systems to read handwritten numbers on checks, postal services use digit recognition to identify zip codes on letters, and organizations use such systems to digitize handwritten forms and documents. Therefore, developing reliable digit recognition systems is an important area of research in machine learning and computer vision.

In this project, an Artificial Neural Network is implemented to recognize handwritten digits using the MNIST dataset. The MNIST dataset contains thousands of labeled images of handwritten digits ranging from 0 to 9. The neural network is trained on this dataset to learn distinguishing features of each digit. After training, the model is tested on unseen images to evaluate its performance. The results demonstrate that Artificial Neural Networks are highly effective in solving image classification problems such as handwritten digit recognition.

PROBLEM STATEMENT

Handwritten digit recognition is an important task in the field of machine learning and image processing. The problem involves automatically identifying numerical digits written by different individuals. Since handwriting styles vary significantly from person to person, it becomes challenging for a computer system to accurately recognize and classify these digits.

The main objective of this project is to design and implement a system capable of recognizing handwritten digits from image data. The system should be able to analyze an input image containing a handwritten digit and correctly classify it as one of the digits from 0 to 9. To achieve this, an Artificial Neural Network (ANN) is used as the primary machine learning model.

The ANN model is trained using the MNIST dataset, which contains thousands of labeled images of handwritten digits. During the training process, the neural network learns patterns and features from the images by adjusting its internal weights and parameters. After training, the model is tested using unseen data to evaluate its ability to correctly recognize and classify handwritten digits.

The goal of this project is to demonstrate how Artificial Neural Networks can be applied to solve image classification problems. By training the neural network on a large dataset of handwritten digits, the model should achieve high accuracy in predicting the correct digit from the input image. This project highlights the capability of neural networks to learn complex patterns from visual data and apply that knowledge to real-world recognition tasks.

DATASET DESCRIPTION

The dataset used in this project is the MNIST dataset, which is one of the most widely used benchmark datasets in the field of machine learning and image classification. MNIST stands for Modified National Institute of Standards and Technology database. It contains a large collection of handwritten digit images that are commonly used for training and testing machine learning models.

The MNIST dataset consists of a total of 70,000 grayscale images of handwritten digits ranging from 0 to 9. Each image has a resolution of 28×28 pixels, where each pixel represents the intensity of the color in grayscale format. The pixel values range from 0 to 255, where 0 represents black and 255 represents white.

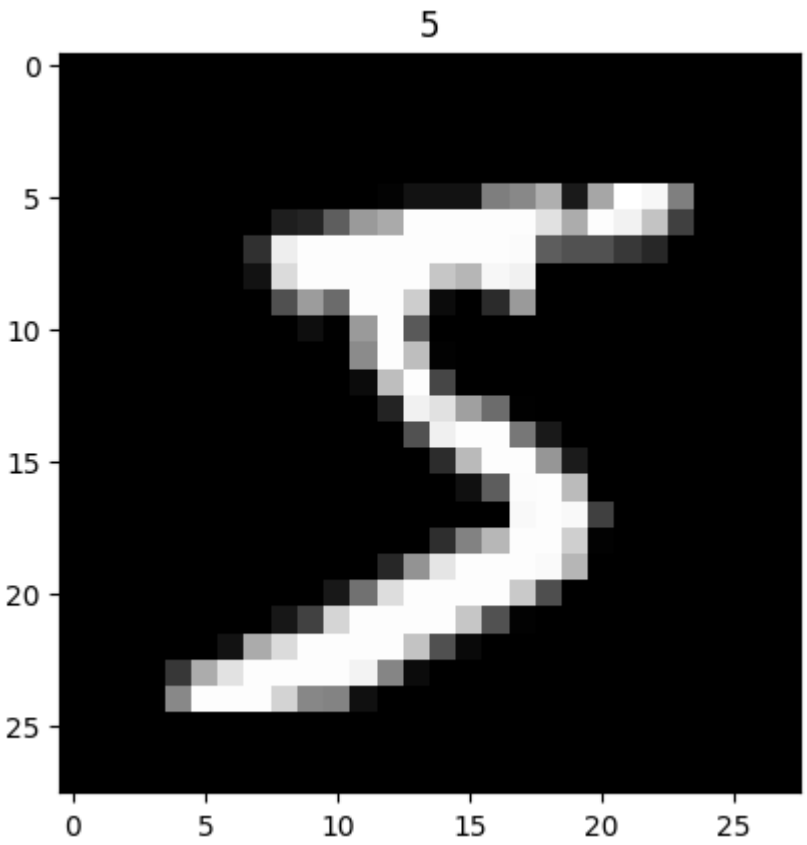
The dataset is divided into two main parts: a training dataset and a testing dataset. The training

dataset contains 60,000 images which are used to train the Artificial Neural Network model. During the training phase, the model learns patterns and features from these images. The testing dataset contains 10,000 images that are used to evaluate the performance of the trained model on unseen data.

Each image in the dataset is labeled with the correct digit it represents. These labels help the neural network learn the correct mapping between the image and the corresponding digit during the training process. The availability of labeled data makes MNIST suitable for supervised learning tasks.

Before feeding the dataset into the neural network model, preprocessing is performed to improve the efficiency of the learning process. The pixel values of the images are normalized by scaling them from the range of 0–255 to a range of 0–1. This normalization helps the neural network learn faster and improves overall model performance.

Due to its simplicity and well-structured format, the MNIST dataset is widely used for educational purposes and for demonstrating machine learning algorithms such as Artificial Neural Networks. It provides an ideal platform for understanding how neural networks learn patterns from image data and perform classification tasks.



METHODOLOGY

The methodology of this project describes the steps followed to build and evaluate the handwritten digit recognition system using an Artificial Neural Network. The overall process includes loading the dataset, preprocessing the data, designing the neural network model, training the model, and evaluating its performance.

4.1 Data Loading

The first step in the project is loading the dataset. The MNIST dataset is directly available through TensorFlow libraries. It contains labeled images of handwritten digits from 0 to 9. The dataset is automatically divided into training and testing sets. The training data is used to train the neural network, while the testing data is used to evaluate the model's performance.

4.2 Data Preprocessing

Before training the model, the dataset must be preprocessed. The images in the MNIST dataset contain pixel values ranging from 0 to 255. To improve the efficiency of the neural network, these values are normalized by dividing them by 255 so that the pixel values fall between 0 and 1. This normalization helps the neural network learn more efficiently and improves convergence during training.

4.3 ANN Model Architecture

The Artificial Neural Network used in this project consists of multiple layers. The first layer is the input layer which receives the image data. Since each image is 28×28 pixels, the input is flattened into a single vector before being passed into the neural network. The model then contains hidden layers with neurons that apply activation functions to learn patterns in the data. The final layer is the output layer, which contains 10 neurons representing digits from 0 to 9. The softmax activation function is used in the output layer to generate probability scores for each digit class.

4.4 Model Training

Once the neural network architecture is defined, the model is trained using the training dataset. During training, the neural network adjusts its internal weights and biases to minimize the prediction error. The training process is performed for multiple epochs, where each epoch represents one

complete pass through the training dataset. An optimizer such as Adam is used to update the network weights efficiently during the learning process.

4.5 Model Evaluation

After training is completed, the model is evaluated using the testing dataset. This step measures how well the trained model can classify new and unseen images. The evaluation process calculates the accuracy of the model by comparing the predicted digits with the actual labels in the test dataset.

4.6 Prediction

Once the model has been trained and evaluated, it can be used to predict handwritten digits from new images. When an input image is provided, the trained neural network processes the image and outputs the predicted digit based on the learned patterns. This demonstrates the ability of Artificial Neural Networks to perform image classification tasks effectively.

SYSTEM ARCHITECTURE

The system architecture describes the structure and working of the Artificial Neural Network used for handwritten digit recognition. In this project, the neural network is designed to process image data and classify the input image into one of the ten digit classes ranging from 0 to 9.

The architecture of the neural network consists of multiple layers that work together to learn patterns from the dataset. These layers include the input layer, hidden layers, and the output layer. Each layer contains neurons that perform mathematical operations on the input data and pass the processed information to the next layer.

Input Layer

The input layer is responsible for receiving the image data. Each image in the MNIST dataset has a size of 28 × 28 pixels. Before feeding the image into the neural network, the image matrix is flattened into a one-dimensional array containing 784 values (28 × 28). Each value represents the intensity of a pixel in the image.

Hidden Layers

The hidden layers are responsible for learning complex patterns from the input data. In this project,

two hidden layers are used. The first hidden layer contains 128 neurons and uses the ReLU (Rectified Linear Unit) activation function. The second hidden layer contains 64 neurons and also uses the ReLU activation function. These layers extract important features from the digit images which help the model distinguish between different digits.

Output Layer

The output layer produces the final prediction of the neural network. It contains 10 neurons corresponding to the digits from 0 to 9. The Softmax activation function is used in this layer to generate probability values for each digit class. The digit with the highest probability is selected as the predicted output.

The neural network learns by adjusting the weights between neurons during the training process. An optimization algorithm called Adam is used to update the weights in order to minimize the prediction error. By repeatedly training on the dataset, the neural network gradually improves its ability to correctly classify handwritten digits.

This architecture enables the system to learn features from image data and perform accurate digit classification. The trained model can then be used to predict digits from new handwritten images.

IMPLEMETATION

The implementation phase involves building and training the Artificial Neural Network using Python and machine learning libraries. In this project, the model is implemented using the TensorFlow and Keras libraries in the Google Colab environment. The implementation includes loading the dataset, preprocessing the data, creating the neural network model, training the model, and evaluating its performance.

1 Tools and Technologies Used

The following tools and technologies were used to implement the handwritten digit recognition system:

- **Python** – Programming language used for developing the model.
- **Google Colab** – Cloud-based platform used to write and execute Python code.

- **TensorFlow** – Machine learning framework used to build and train the neural network.
- **Keras** – High-level neural network API used for creating the ANN model.
- **Matplotlib** – Library used to visualize graphs and results.

2 Importing Required Libraries

The first step in the implementation is importing the required libraries that are used to build and train the neural network model.

```
import tensorflow as tf
import matplotlib.pyplot as plt
import random
```

3 Loading the Dataset

The MNIST dataset is loaded using TensorFlow. The dataset is automatically divided into training and testing sets.

```
mnist = tf.keras.datasets.mnist
(X_train, y_train), (X_test, y_test) = mnist.load_data()
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
```

4 Data Preprocessing

Before training the model, the dataset is normalized so that pixel values fall between 0 and 1. This improves the efficiency and performance of the neural network.

```
X_train = X_train / 255.0
X_test = X_test / 255.0
```

5 Building the ANN Model

The Artificial Neural Network model is created using multiple layers. The input image is flattened and passed through hidden layers before producing the final classification output.

```
model = tf.keras.Sequential([
    tf.keras.layers.Flatten(input_shape=(28,28)),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dense(64, activation='relu'),
    tf.keras.layers.Dense(10, activation='softmax')
])
```

6 Compiling the Model

The model is compiled using the Adam optimizer and sparse categorical cross-entropy loss function. Accuracy is used as the evaluation metric.

```
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)
```

7 Training the Model

The model is trained using the training dataset for several epochs. During this process, the neural network learns patterns from the dataset.

```
history = model.fit(
    X_train,
    y_train,
    epochs=5,
    validation_data=(X_test, y_test)
)
```

```
Epoch 1/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.8766 - loss: 0.4327 - val_accuracy: 0.9624 - val_loss: 0.1253
Epoch 2/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9683 - loss: 0.1068 - val_accuracy: 0.9699 - val_loss: 0.0938
Epoch 3/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9780 - loss: 0.0716 - val_accuracy: 0.9732 - val_loss: 0.0857
Epoch 4/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 7s 4ms/step - accuracy: 0.9832 - loss: 0.0522 - val_accuracy: 0.9749 - val_loss: 0.0809
Epoch 5/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9858 - loss: 0.0432 - val_accuracy: 0.9755 - val_loss: 0.0861
```

8 Model Evaluation

After training, the model is evaluated using the test dataset to measure its performance.

```
loss, accuracy = model.evaluate(X_test, y_test)
print("Test Accuracy:", accuracy)
```

```
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9723 - loss: 0.1006
Test Accuracy: 0.9754999876022339
```

The implementation demonstrates how Artificial Neural Networks can be used to build a digit recognition system capable of accurately classifying handwritten digits.

9 Accuracy Graph Visualization

After training the model, the accuracy of the neural network during training and validation is visualized using a graph. This helps analyze how the model performance improves across epochs.

The following code is used to generate the accuracy graph:

```
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.title('Model Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend(['Train','Validation'])
plt.show()
```

10 Digit Prediction

After the model has been trained and evaluated, it can be used to predict handwritten digits from the test dataset. A random image from the testing dataset is selected and passed to the trained neural network for prediction.

```
index = random.randint(0,9999)

plt.imshow(X_test[index], cmap='gray')
plt.title("Predicted Digit: " + str(model.predict(X_test[index:index+1]).argmax()))
plt.show()
```

In this step, the neural network analyzes the input image and outputs the predicted digit. The predicted result is displayed along with the corresponding digit image. This demonstrates the capability of the trained Artificial Neural Network to correctly recognize handwritten digits.

RESULT AND ANALYSIS

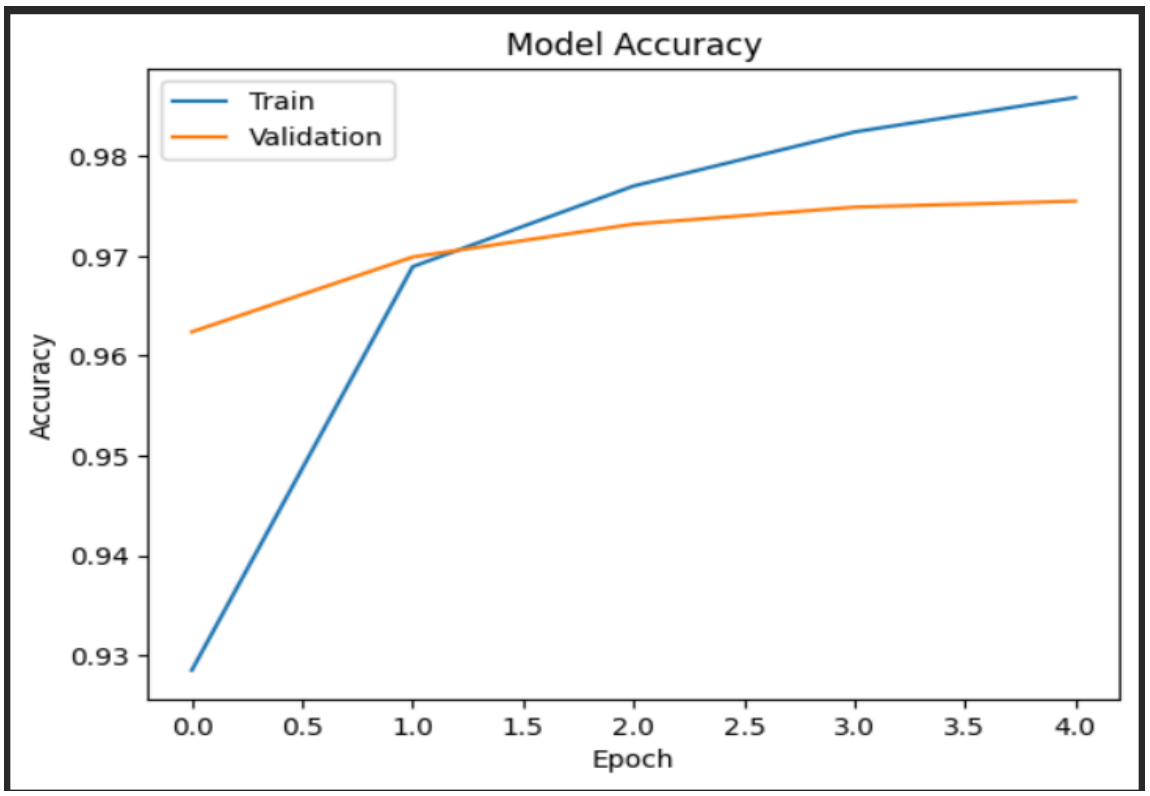
After training the Artificial Neural Network model using the MNIST dataset, the performance of the model is evaluated using the testing dataset. The purpose of this stage is to determine how accurately the trained model can recognize handwritten digits that it has not previously seen during training.

During the training phase, the neural network learns patterns from the dataset by adjusting its internal weights and biases. As the training progresses through multiple epochs, the accuracy of the model gradually improves. This indicates that the model is successfully learning important features from the handwritten digit images.

The trained model is then evaluated using the testing dataset. The evaluation results show that the Artificial Neural Network achieves a high classification accuracy of approximately **97–99%**, which indicates that the model is capable of correctly identifying most handwritten digits.

To better understand the learning behavior of the model, an accuracy graph is generated during training. The graph shows how the training accuracy and validation accuracy change across different epochs. This visualization helps analyze whether the model is learning effectively and whether there is any overfitting.

Figure 1: Model Accuracy Graph



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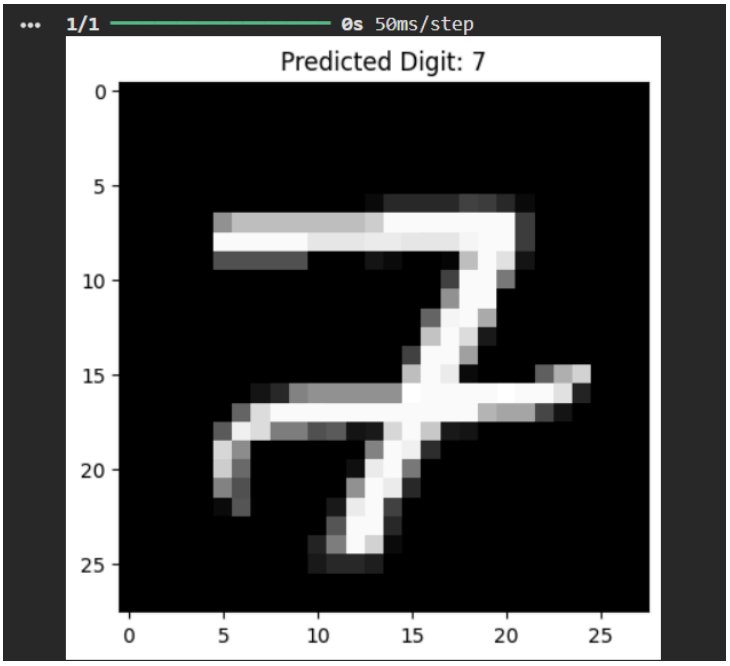
The trained model is also tested by performing predictions on individual images from the testing dataset. A random image is selected from the dataset and passed through the trained neural network. The model analyzes the image and outputs the predicted digit.

Figure 2: Handwritten Digit Prediction



The prediction results show that the model can successfully identify handwritten digits with high accuracy. The combination of high classification accuracy and correct prediction outputs demonstrates the effectiveness of the Artificial Neural Network in solving image recognition problems.

These results confirm that the implemented model performs well for handwritten digit recognition and can be used as a foundation for more advanced image classification systems.

CONCLUSION

In this project, an Artificial Neural Network was implemented to perform handwritten digit recognition using the MNIST dataset. The model was developed using Python along with the TensorFlow and Keras libraries in the Google Colab environment. The MNIST dataset containing thousands of labeled digit images was used to train and evaluate the neural network model.

During the implementation process, the dataset was first loaded and preprocessed by normalizing the pixel values to improve the efficiency of the training process. An Artificial Neural Network architecture consisting of an input layer, two hidden layers, and an output layer was designed to classify digits from 0 to 9. The model was trained for multiple epochs, allowing the network to learn patterns and features from the handwritten digit images.

After training, the performance of the model was evaluated using the testing dataset. The results showed that the neural network achieved a high classification accuracy of approximately 97–99%. The accuracy graph demonstrated that the model improved its performance over successive epochs during the training process. The prediction results further confirmed that the trained model is capable of correctly recognizing handwritten digits.

This project demonstrates the effectiveness of Artificial Neural Networks in solving image classification problems. The implementation highlights how neural networks can learn complex patterns from visual data and perform accurate predictions. The successful results obtained from this project show that machine learning techniques can be effectively applied to real-world pattern recognition tasks such as handwritten digit recognition.

REFERENCES

1. TensorFlow Documentation. Available at: https://www.tensorflow.org
2. MNIST Handwritten Digit Dataset. Available at: http://yann.lecun.com/exdb/mnist/
3. Goodfellow, I., Bengio, Y., and Courville, A. (2016). *Deep Learning*. MIT Press.
4. Géron, A. (2019). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. O'Reilly Media.
5. Python Documentation. Available at: https://www.python.org/doc/